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Assignment No: 5

Problem Statement:

Generate creative artwork using Stable Diffusion.

Objective:

- To understand the Stable Diffusion architecture and how diffusion models synthesise images from text prompts.
- To load and configure Stable Diffusion v1.5 with float16 precision on a GPU using Hugging Face Diffusers.
- To design an effective text prompt and understand how prompt engineering influences image quality and style.
- To generate a 512x512 image via the full latent diffusion pipeline and display it within Colab.
- To save the generated artwork as a PNG file for downstream use.

S/W Packages and H/W apparatus used:

- Operating System: Windows/Linux/macOS
- Kernel: Python 3.x
- Tools: Google Colab (GPU Runtime - T4)
- Hardware: NVIDIA T4 GPU — GPU mandatory for practical inference speeds
- Libraries and packages used: Diffusers (StableDiffusionPipeline), Transformers, Accelerate, PyTorch (torch.float16)

Theory:

Stable Diffusion is a latent text-to-image diffusion model developed by Stability AI, CompVis, and RunwayML. It can produce photorealistic and stylised images from free-form text descriptions. Unlike earlier pixel-space diffusion systems, Stable Diffusion performs its diffusion process in a compressed latent space, dramatically lowering memory and compute requirements while preserving high visual fidelity.

Diffusion Models:

Diffusion models are generative models that learn to reverse a forward noise-addition process. During training the model learns to predict and remove noise across T timesteps. At inference, starting from pure Gaussian noise, the model iteratively denoises over T steps guided by the text condition, progressively recovering a coherent image. The denoising backbone is a UNet neural network.

Latent Diffusion Model (LDM):

Instead of operating on full-resolution pixel tensors, Stable Diffusion uses a Variational Autoencoder (VAE) to compress images into a latent space roughly 8x smaller (e.g., 64x64x4 instead of 512x512x3). The UNet denoises within this compact space, and the VAE decoder reconstructs the final pixel image from the cleaned latent. This design drastically reduces computation while maintaining visual quality.

CLIP Encoder and Cross-Attention:

Text conditioning is provided by a CLIP text encoder that converts the prompt into embeddings injected into the UNet through cross-attention at multiple layers. Classifier-free guidance (CFG) amplifies the text conditioning signal by taking a weighted difference between conditional and unconditional denoising directions, with a guidance scale around 7.5 controlling the fidelity-diversity trade-off.

Prompt Engineering:

Generated image quality is highly sensitive to prompt construction. Effective prompts combine a clear subject, artistic style modifiers (e.g., 'digital art', 'oil painting'), quality boosters (e.g., 'highly detailed', '8K'), and mood descriptors. The prompt 'a futuristic city floating in the sky, digital art, highly detailed' combines all these elements to guide the model towards a visually rich, coherent scene.

Methodology:

- Install diffusers, transformers, accelerate, and torch.
- Import torch and StableDiffusionPipeline from diffusers.
- Load runwayml/stable-diffusion-v1-5 with torch_dtype=torch.float16 for half-precision GPU inference.
- Transfer the pipeline to the CUDA device via pipe.to('cuda').
- Define the prompt: 'a futuristic city floating in the sky, digital art, highly detailed'.
- Run pipe(prompt).images[0] — CLIP encoding, 50-step UNet denoising, VAE decoding — to obtain a PIL image.
- Display the image inline in the notebook.
- Save the output as 'creative_art.png' via image.save().

Use Cases:

- Use Case 1 - Concept Art and Rapid Prototyping: Designers and game studios generate environment, character, and prop concept art from text descriptions, accelerating early ideation without manual sketching.
- Use Case 2 - Marketing Visual Content: Brands create custom campaign visuals and product mockups from branded style prompts, reducing reliance on stock libraries and graphic designers.
- Use Case 3 - Educational Illustration: Educators generate custom diagrams, historical reconstructions, and concept visualisations from descriptive prompts, producing engaging, copyright-free teaching materials.

Advantages:

- Generates photorealistic and artistically diverse images from text alone, requiring no drawing ability.
- float16 inference halves GPU memory consumption, fitting the model within a 16GB T4 GPU.
- The Diffusers pipeline bundles tokenization, encoding, denoising, and decoding into one callable.
- PNG outputs are immediately usable in downstream creative or digital applications.
- Wide artistic style range is achievable through prompt modifiers without any retraining.

Limitations:

- GPU with at least 6-8GB VRAM is required; CPU inference takes hours per image.

- Precise spatial relationships, accurate text in images, and human anatomy are common failure modes.
- Generation is non-deterministic — identical prompts produce different images without a fixed seed.
- Images above 512x512 demand substantially more VRAM and generation time.
- The model may reflect gender, racial, and cultural biases present in its LAION training dataset.

Applications:

- Game and film pre-production: Rapid concept art, character designs, and texture references.
- Fashion and product design: Visual mockups of clothing concepts and packaging artwork.
- Architecture and interior visualisation: Rendering building facades and room layouts from text.
- Book and comic illustration: Custom character portraits and scene compositions.
- Scientific visualisation: Artistic representations of molecular structures, astronomical phenomena, and quantum effects.

Conclusion:

This assignment demonstrated AI-driven creative artwork generation using Stable Diffusion v1.5 via Hugging Face Diffusers. The pipeline was loaded in float16 precision on a T4 GPU and the prompt 'a futuristic city floating in the sky, digital art, highly detailed' was processed through CLIP text encoding, 50-step UNet denoising in latent space, and VAE decoding to produce a 512x512 image. The artwork was displayed and saved as a PNG. The experiment illustrates how state-of-the-art generative AI enables prompt-driven image synthesis, unlocking transformative capabilities across creative industries, education, and design with minimal technical overhead.